Project summary

Project	PRUEBA01
Run ID	RUN001
Technology	Illumina
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Advisor	Nadia Queralt Ortega
Contact	nadiiaqueraltt@gmail.com

Introduction

The aim of this document is to drive you through the reading of the analysis performed by the FISABIO's Bioinformatics Service, providing a user-friendly approach to metagenomics.

By clicking on hyperlinks it is possible to navigate among the table of contents, figures, tables and bibliographic references.

All generated figures can be found in the `images/` folder. Each file name self-explains the data. Figures are provided in PNG (Portable Network Graphic) format. For direct editing of the PNG images we suggest the use of the Open Source software Inkscape.

Tables are formatted by tabs producing standard `.tsv` files. All tables can be easily imported in almost every spreadsheet software.

If a file is very big in size, we store it in the `gzip` format, a common unix compression algorithm. It can be easily unzipped with:

```
# Decompress a single gzip file
gunzip file.gz

# Compress a file
gzip file
```

Another kind of file which includes a tree folder structure is labeled by the extension `.tar.gz` and can be decompressed using:

```
# only unix systems
tar xvzf file.tar.gz
```

In **Microsoft Windows** systems, standard decompressing tools should be able to access these kinds of files.

We hope you find this report helpful and for any suggestion or bug notification, please contact us by e-mail to: nadiiaqueraltt@gmail.com.

Data analysis

Samples have been co-assembled to maximise contig construction robustness. Contigs were annotated for functional and taxonomic affiliations (binning), and individual samples were then mapped against the assembled contigs to calculate taxonomic and functional statistics.

Taxonomy distributions

The following table reports the number of reads mapped against their respective annotated contigs, summarised by taxonomy relationships at the phylogenetic rank superkingdom.

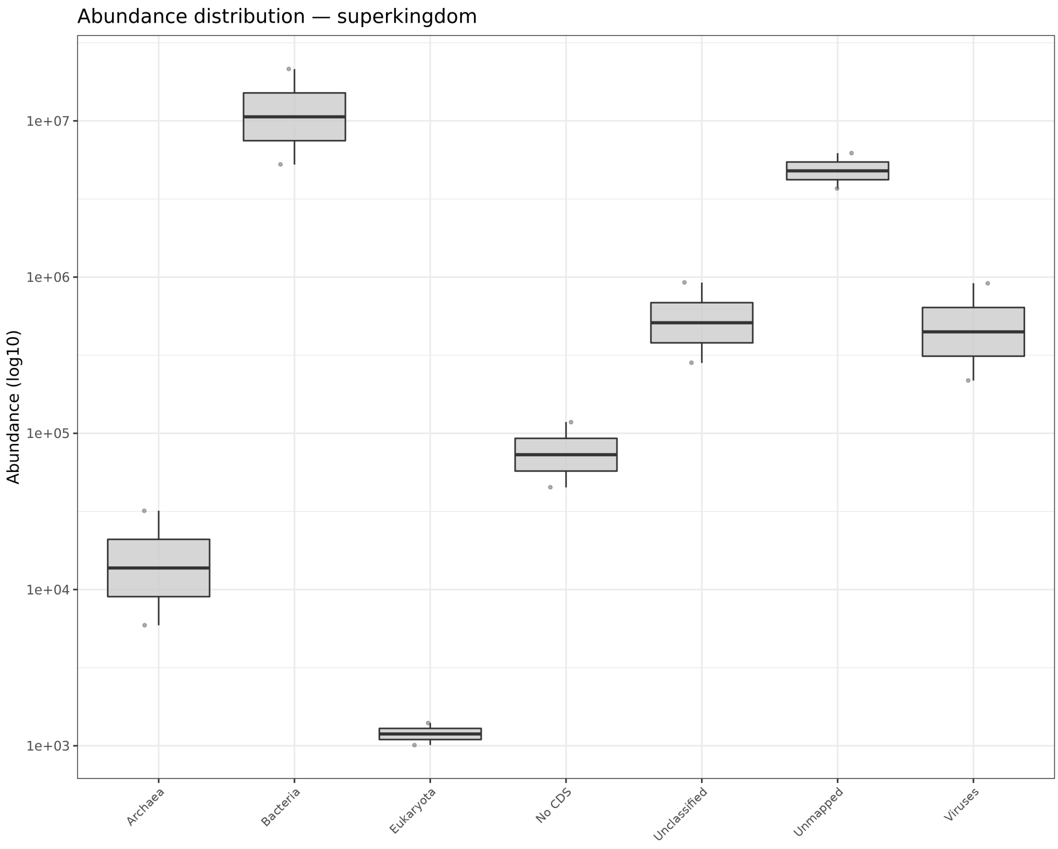
See the superkingdom abundances table in the HTML report.

The following tables report sample taxonomy counts for each taxonomic rank.

File description	Link
Superkingdom abundances table (tsv)	open
Phylum abundances table (tsv)	open
Class abundances table (tsv)	open
Order abundances table (tsv)	open
Family abundances table (tsv)	open
Genus abundances table (tsv)	open
Species abundances table (tsv)	open

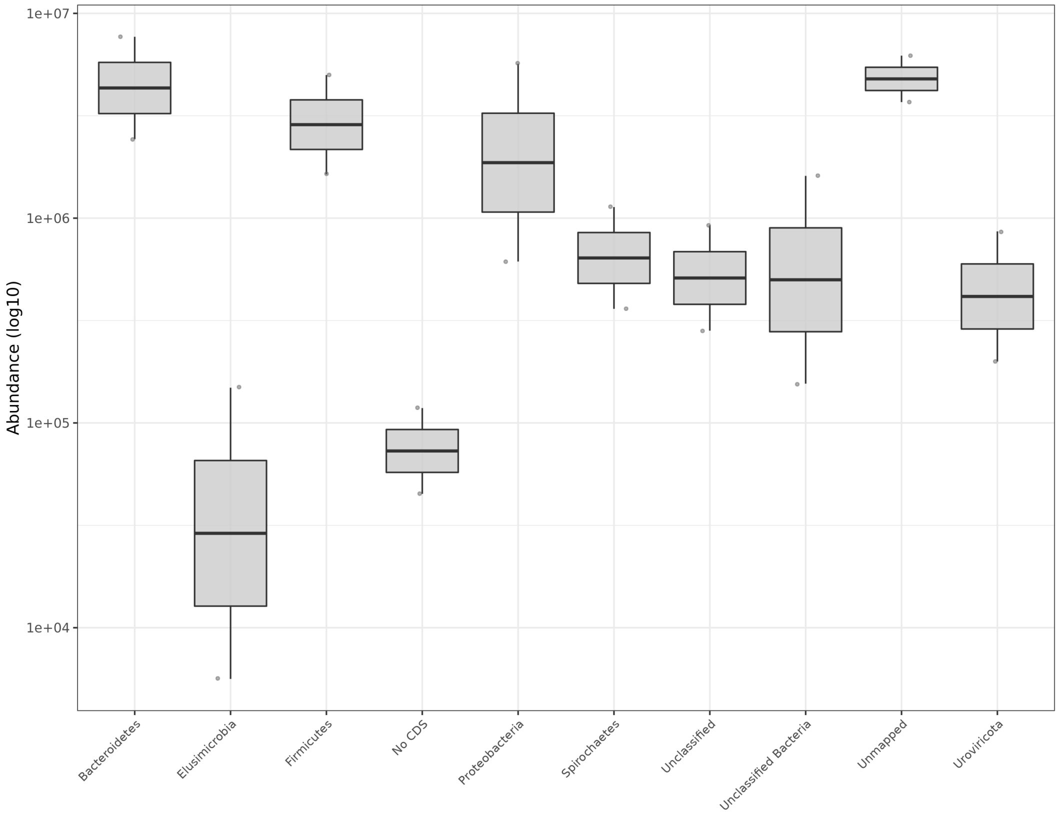
Main taxonomy boxplots

The following figures show a general overview of the top **10** taxa abundance distribution among Kingdom, Phylum, Class, Order, Family, Genus and Species taxonomic ranks.



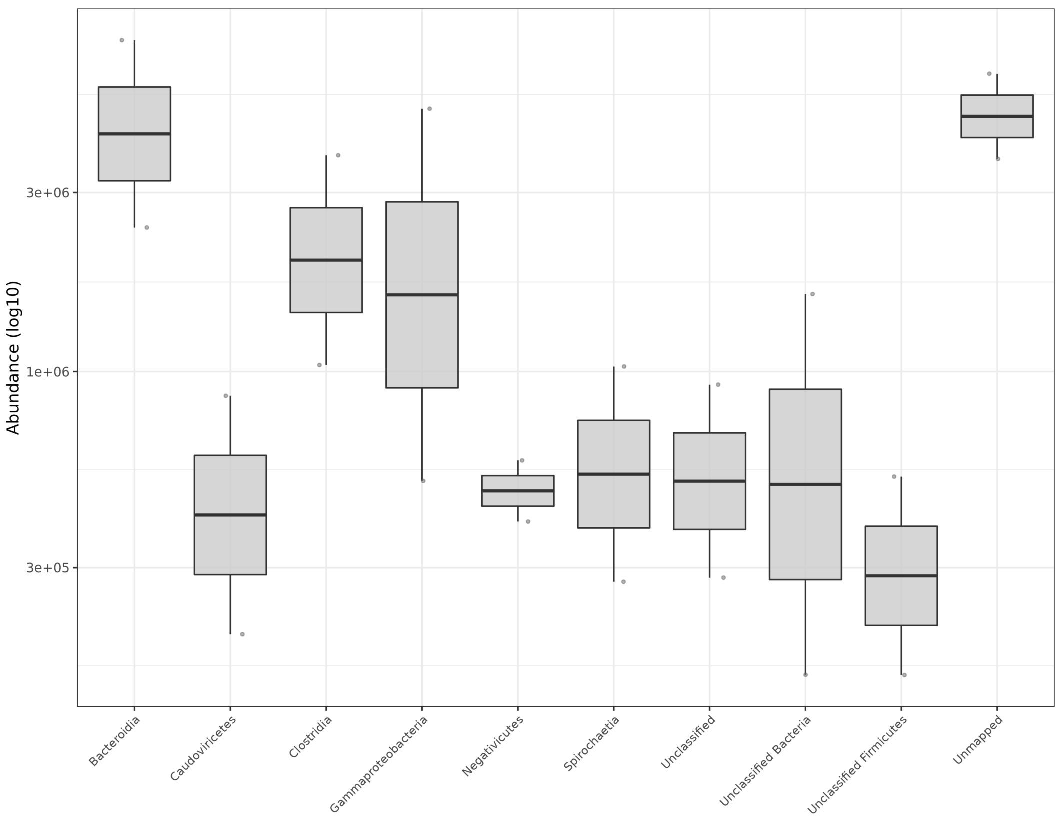
Top 10 taxa among Kingdom taxonomic rank.

Abundance distribution — phylum

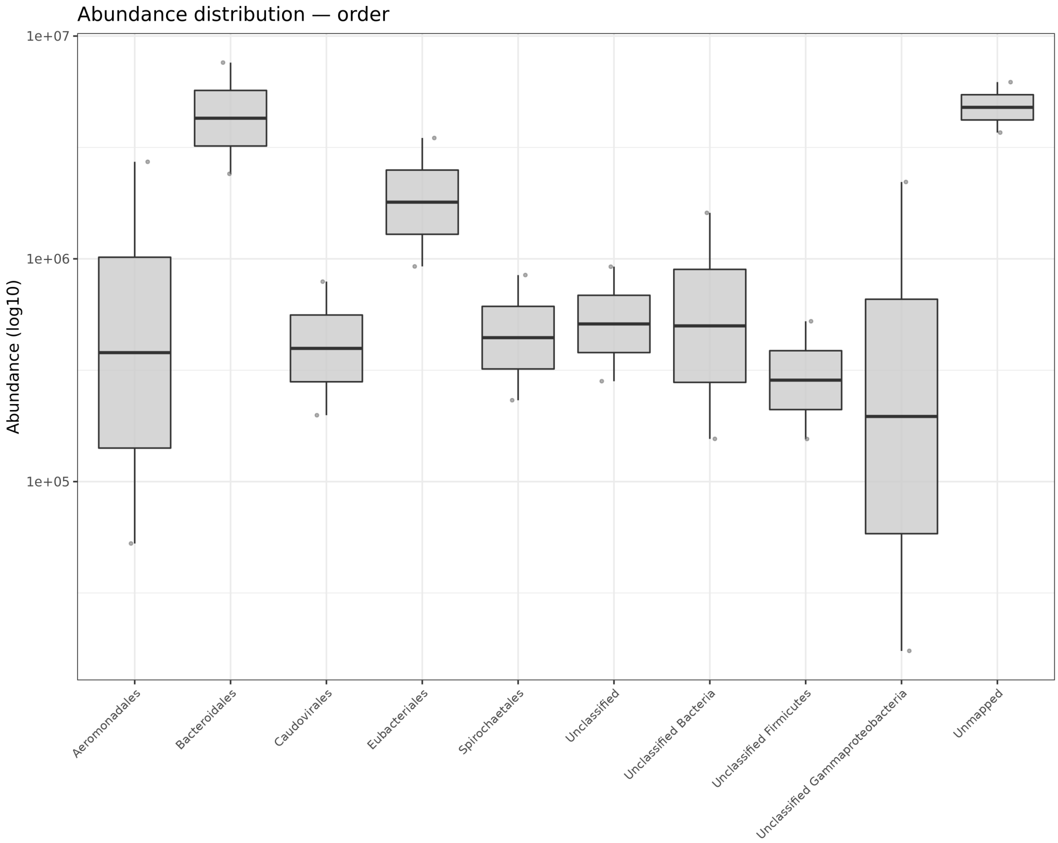


Top 10 taxa among Phylum taxonomic rank.

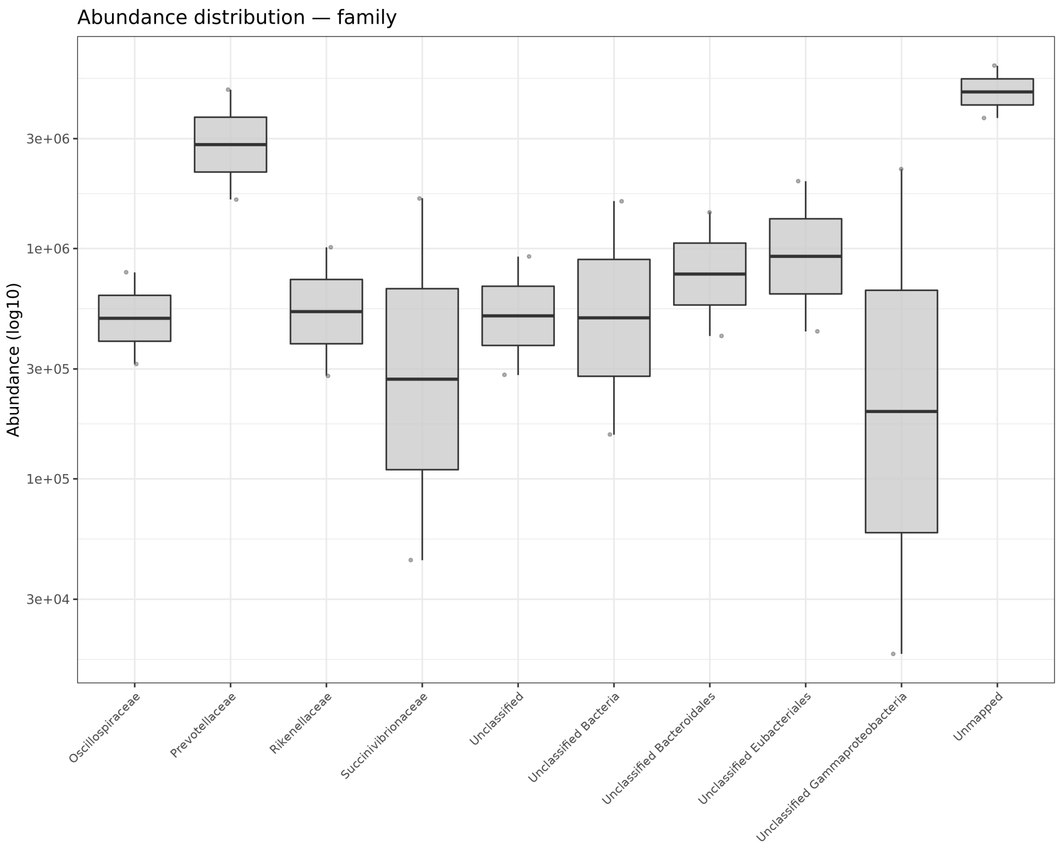
Abundance distribution — class



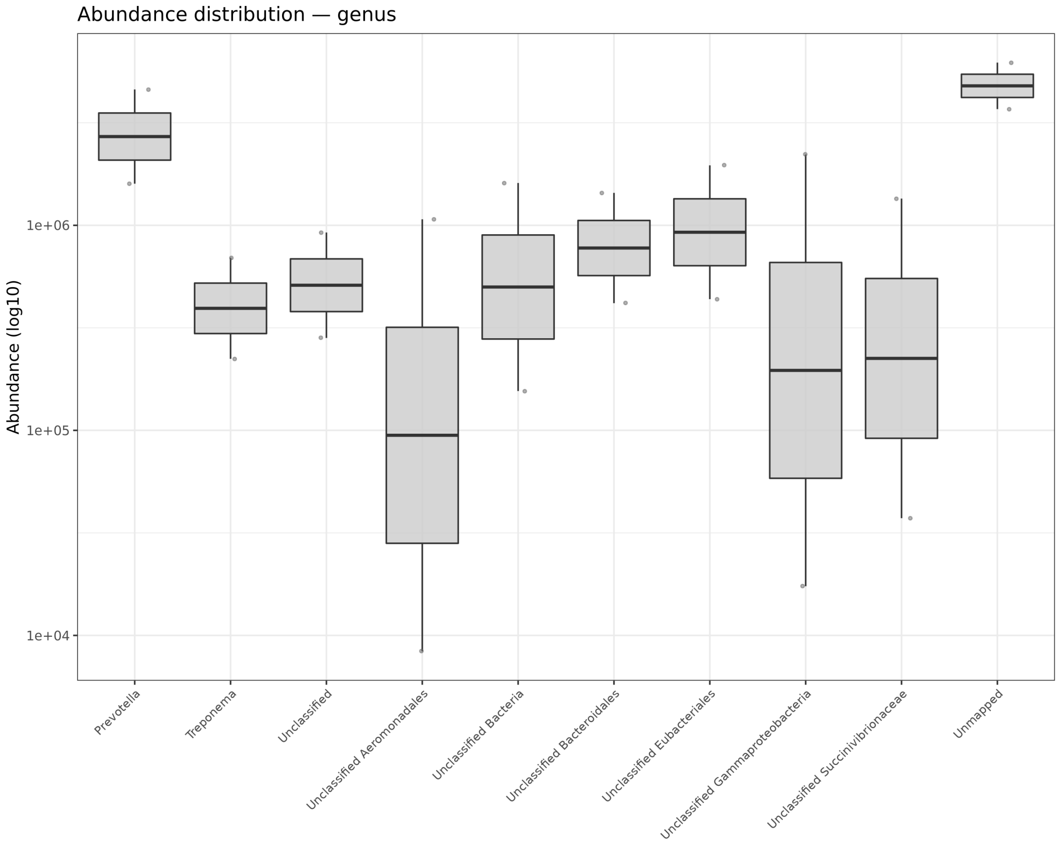
Top 10 taxa among Class taxonomic rank.



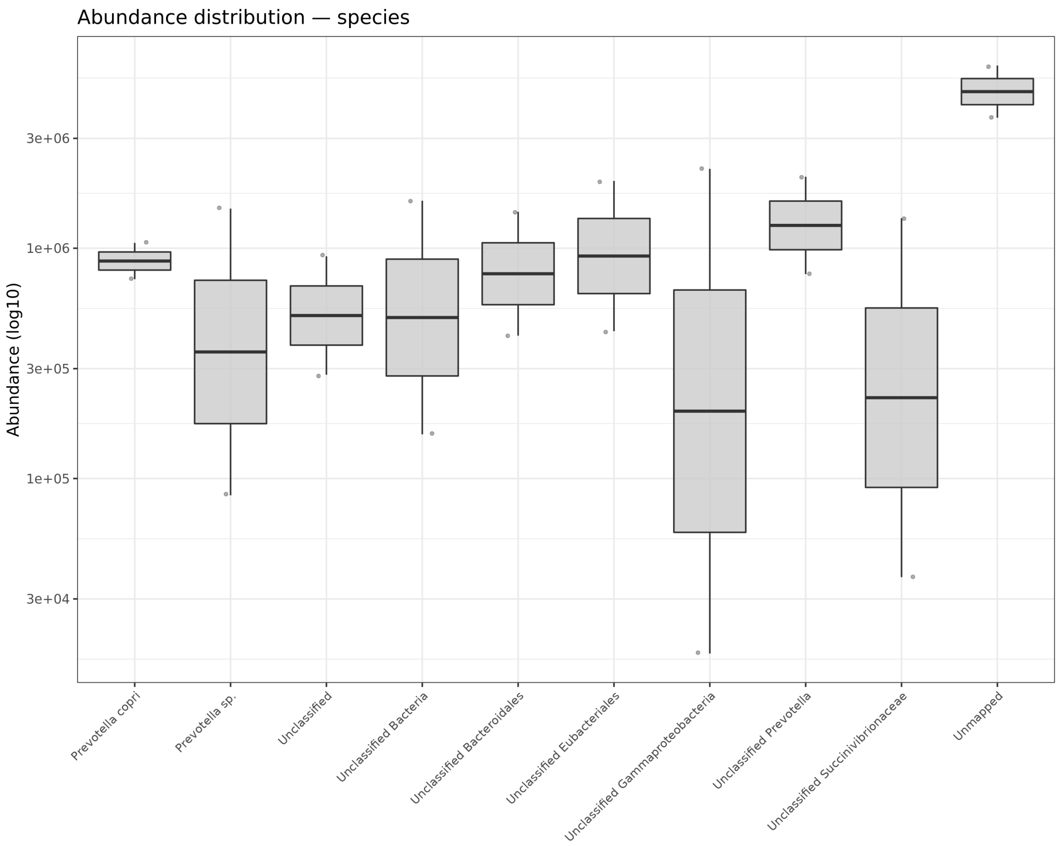
Top 10 taxa among Order taxonomic rank.



Top 10 taxa among Family taxonomic rank.



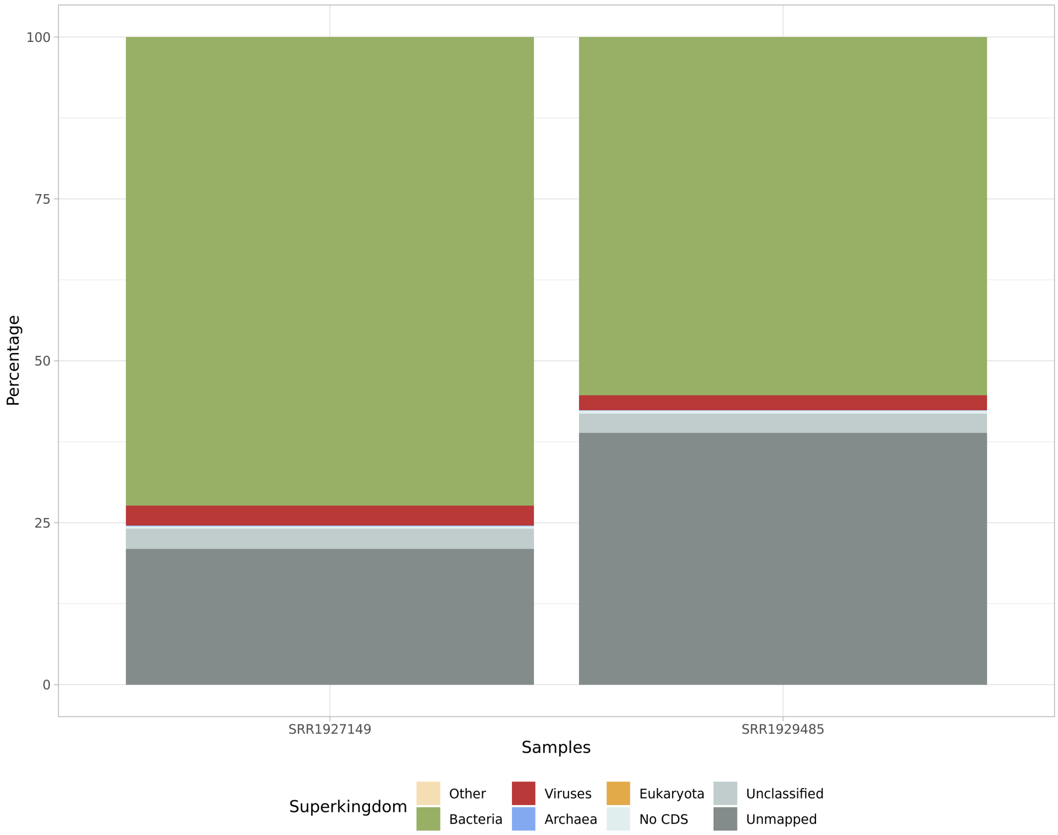
Top 10 taxa among Genus taxonomic rank.



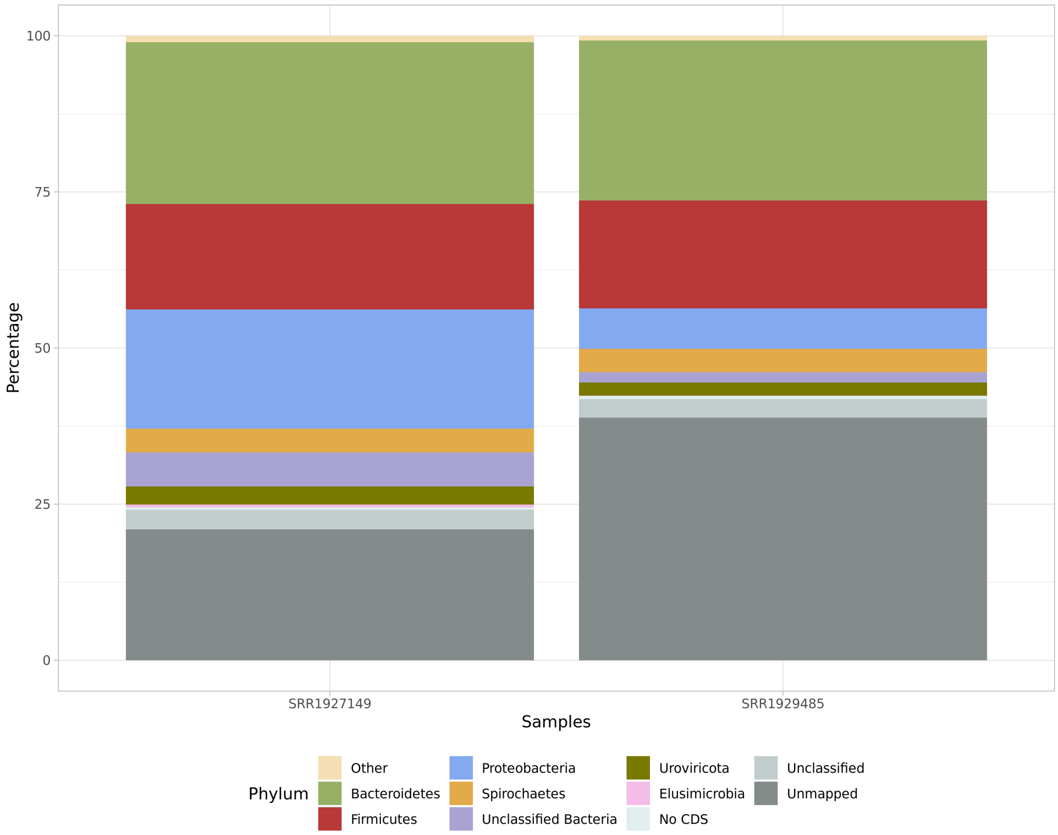
Top 10 taxa among Species taxonomic rank.

Main taxonomy barplots

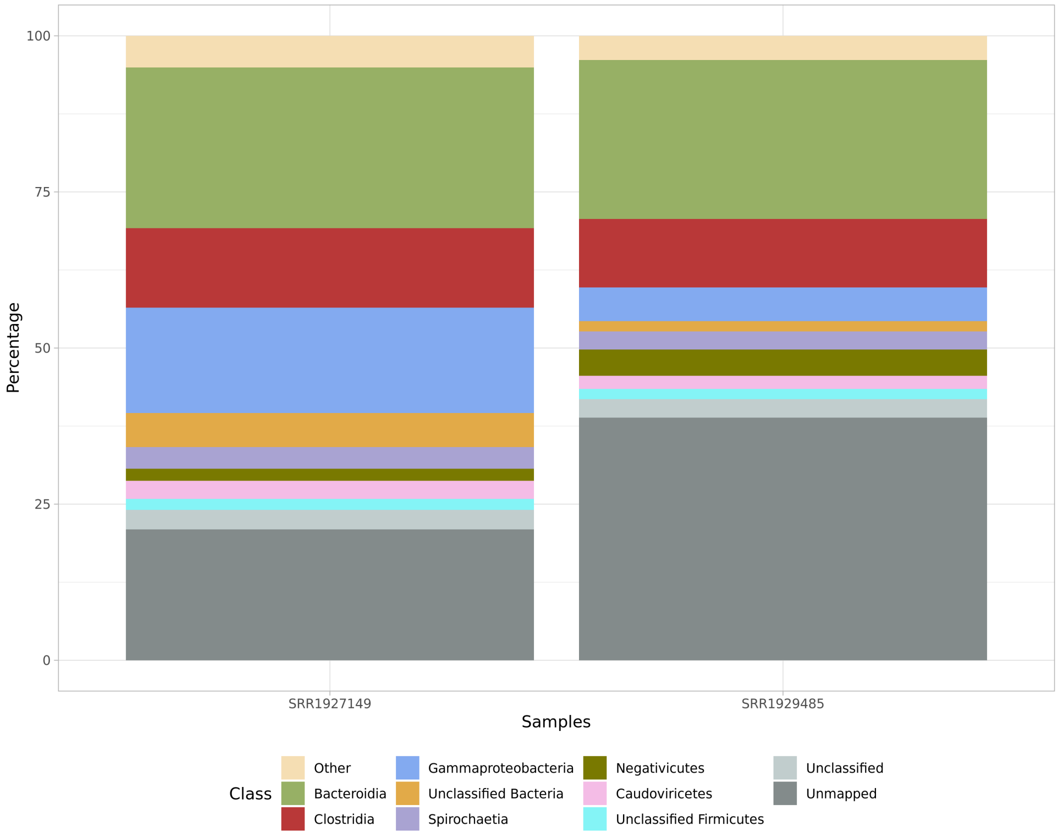
The following figures show the top **10** taxa at each taxonomic rank as percentage barplots.



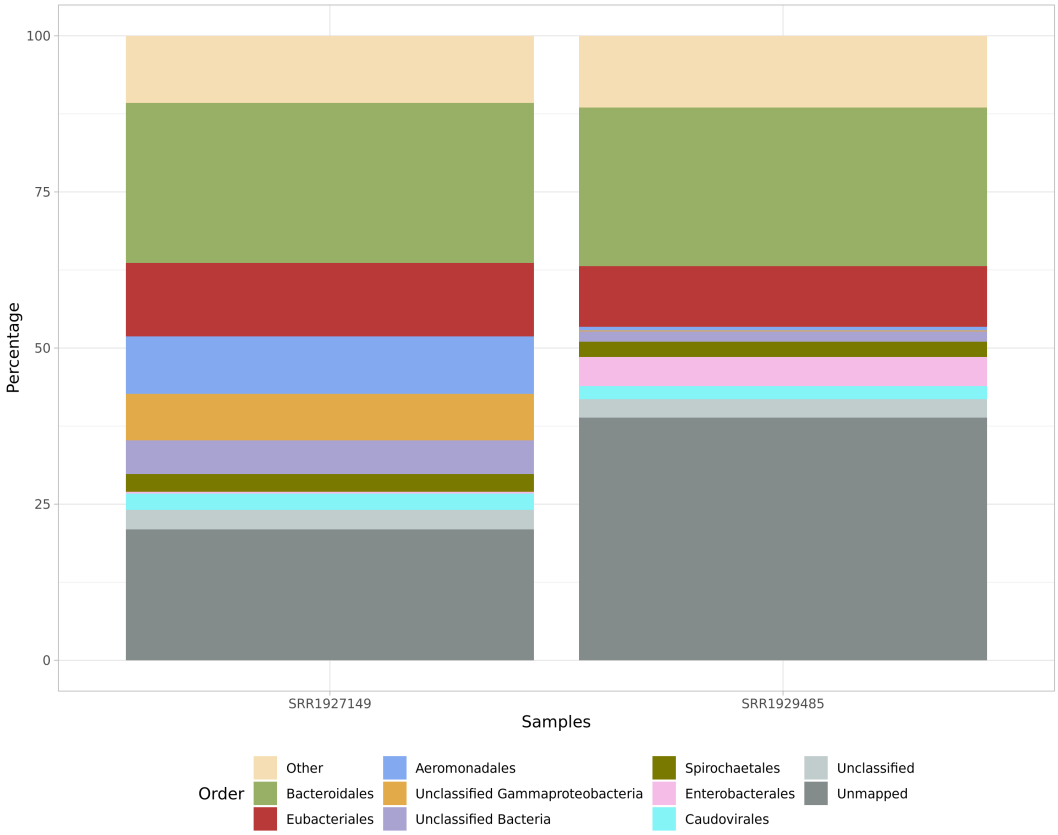
Top 10 taxa at Superkingdom rank.



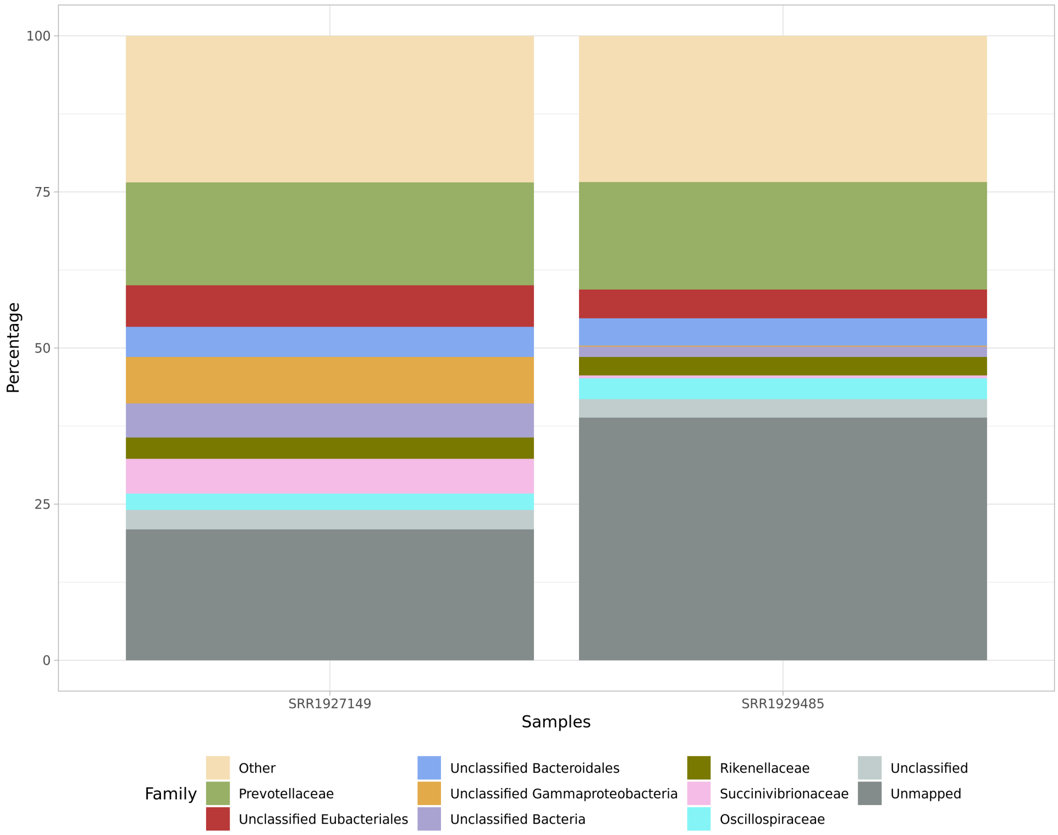
Top 10 taxa at Phylum rank.



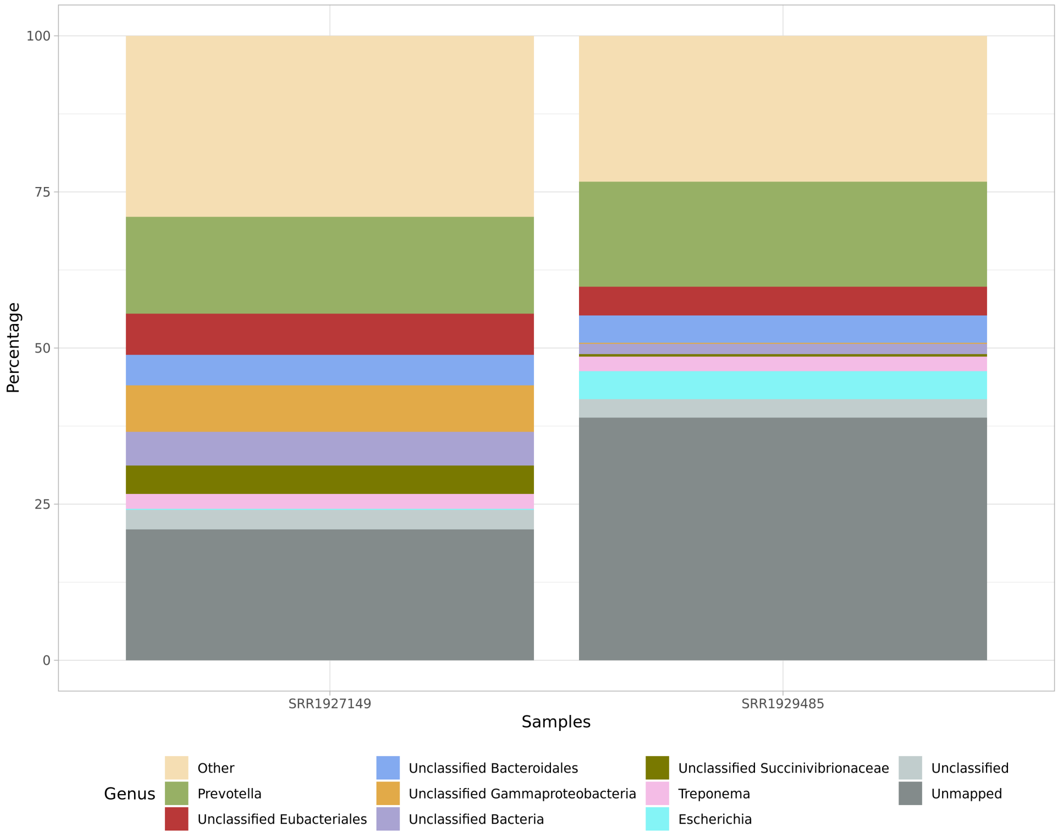
Top 10 taxa at Class rank.



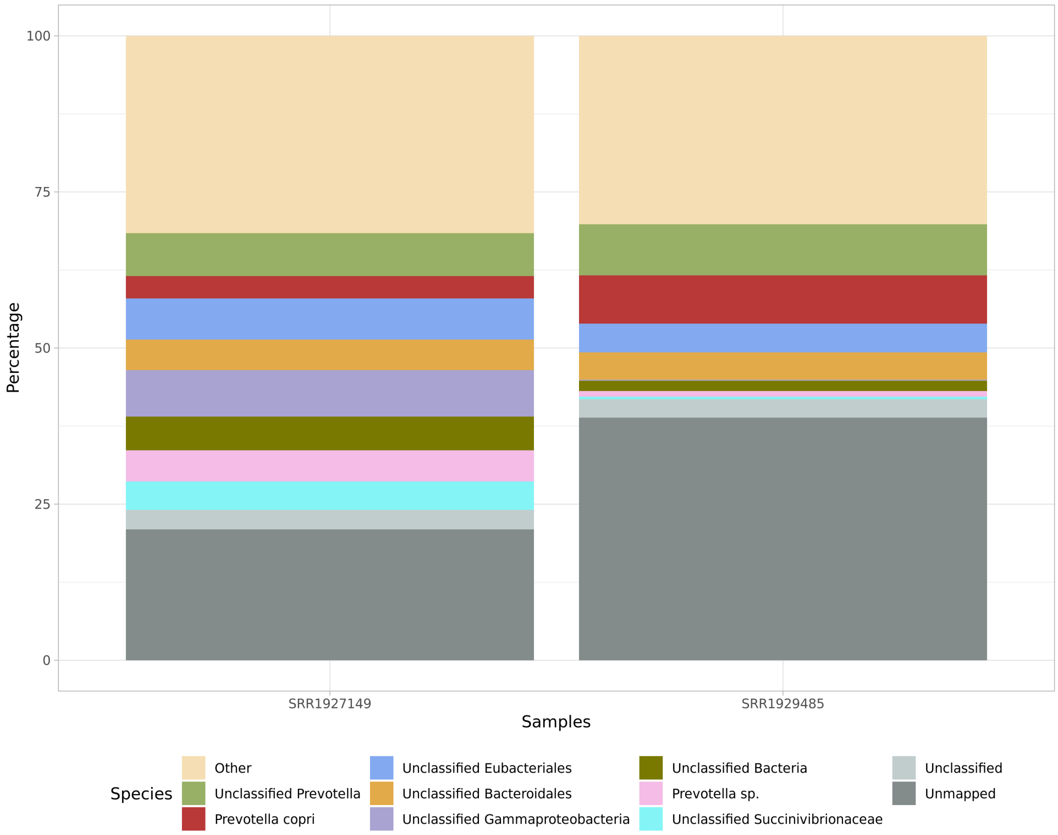
Top 10 taxa at Order rank.



Top 10 taxa at Family rank.



Top 10 taxa at Genus rank.



Top 10 taxa at Species rank.

Interactive taxonomy distribution charts

Binning information from coding domain sequences (CDSs) has been summarised using the KronaViewer format (Ondov et al., 2011).

Open Krona interactive chart

Main functional annotation

File description	Link
Main ORF annotation file	open

KEGG-based annotations and counts

File description	Link
KEGG TPM	open
KEGG Coverage	open
KEGG Raw reads count	open
KEGG Raw bases count	open

COG-based annotations and counts

File description	Link
COG TPM	open
COG Coverage	open
COG Raw reads count	open
COG Raw bases count	open

PFAM-based annotations and counts

File description	Link
PFAM TPM	open
PFAM Coverage	open
PFAM Raw reads count	open
PFAM Raw bases count	open

ORF annotation column description

Column	Description
ORF ID	Unique identifier including contig name and coordinates
Contig ID	Contig identifier (assembler_contig)
Molecule	Annotation type (CDS, tRNA, rRNA, etc.)
Method	Engine used for annotation
Length NT	Nucleotide length of the annotation

Length AA	Amino acid length of the annotation
GC perc	Guanine-Cytosine percentage
Gene name	Gene name when available
Tax	Taxonomy (k_, p_, c_, o_, f_, g_, s_)
KEGG ID	KEGG identifier (e.g. K01953)
KEGGFUN	KEGG function and enzyme code
KEGGPATH	KEGG pathway participation
COG ID	COG identifier (e.g. COG0367)
COGFUN	COG function description
COGPATH	COG pathway participation
PFAM	PFAM identifier and protein function
TPM	Transcripts Per Kilobase Million (one column per sample)
Coverage	Per-sample coverage values
Raw reads count	Reads overlapping annotation per sample
Raw bases count	Bases overlapping annotation per sample

Assembly files

The following files contain auxiliary information about the whole process. You will find, among others, contig sequences and extracted fragments (or complete genes) such as ribosomal sequences, tRNAs, protein coding genes (CDSs) in nucleotide and amino acid versions.

They are stored in **gzipped** format in order to reduce disk consumption. Linux users should be already familiar with this kind of format.

File description	Link
Contigs (gzip compressed FASTA file)	open
Ribosomal sequences (gzip compressed FASTA file)	open
CDSs amino acids (gzip compressed FASTA file)	open
CDSs nucleotides (gzip compressed FASTA file)	open
CDSs coordinates in GFF format (gzip compressed GFF file)	open

Materials and methods

Metagenomics workflow

The metagenomics workflow was carried out using the **SqueezeMeta** pipeline [tamames2018]. This pipeline integrates several bioinformatics tools including assemblers (MEGAHIT [megahit2015], SPAdes [spades2012], Canu [canu2017]), clustering programs (CD-HIT [cdhit2012]), mappers (Bowtie2 [bowtie2012], minimap2 [minimap2018], BWA [bwa2009]), and annotation engines (DIAMOND [diamond2015], Prodigal [prodigal2010], HMMER [hmmer2011], RNAmmer [rnammer2007], Aragorn [aragorn2004], RDP classifier [rdpclassifier2007]), among others.

The protocol employed co-assembly of all reads from all samples into a single assembly (`-m coassembly` SqueezeMeta command). Contigs were searched for CDSs, rRNA, tRNAs, and other genomic features. Each original cleaned dataset was then mapped against the co-assembled contigs, and coverage data were collected per annotation.

SqueezeMeta calculates read abundance, percentage normalisation, and TPM (Transcripts per Kilobase Million) normalisation.

R packages used for the analysis

Several in-house scripts were used to improve the final data description and this report. The pipeline core is run using **R** [rcoreteam]. The **SQMtools** library from SqueezeMeta was employed to extract and reformat the annotation datasets, as well as **ggplot2** [wickham2016] library to save images in .png format.

Pipeline parameters used in this run

```
Project:        PRUEBA01
Run ID:         RUN001
Technology:     Illumina
Top taxa (N):   10
Date:           17 06 2026
```

References

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